European guidelines for topical photodynamic therapy part 1: treatment delivery and current indications – actinic keratoses, Bowen's disease, basal cell carcinoma

Abstract

Topical photodynamic therapy (PDT) is a widely used non-invasive treatment for certain non-melanoma skin cancers, allowing for the treatment of large and multiple lesions with excellent cosmetic outcomes. High efficacy has been demonstrated for PDT using standardized protocols in non-hyperkeratotic actinic keratoses (AK), Bowen's disease (BD), superficial basal cell carcinomas (BCC), and certain thin nodular BCCs, with superior cosmetic results compared to conventional therapies. Recurrence rates following PDT are generally comparable to those of other non-surgical treatments, although they are higher than after surgical excision for nodular BCC. PDT is not recommended for invasive squamous cell carcinoma. The treatment is usually well tolerated, but tingling, discomfort, or pain during illumination is common. Recent studies have identified patient factors associated with increased pain, enabling earlier implementation of pain-minimization strategies. Reduced discomfort has been observed with newer protocols, including shorter photosensitizer application times and daylight PDT for actinic keratoses.

Introduction

These guidelines aim to promote safe and effective practice across Europe for the delivery of topical photodynamic therapy (PDT) in dermatological conditions. They are based on evidence from a systematic literature review (using MEDLINE) and previous clinical guidelines. A companion paper (Part II) reviews emerging and novel indications for topical PDT.

To date, topical PDT has been approved for the treatment of selected non-melanoma skin cancers (NMSC). Currently, only three photosensitizing agents are licensed for use in Europe (see Table 1):

- **Methyl aminolaevulinate (MAL)** – marketed as Metvix®/Metvixia® (Galerma, Lausanne, Switzerland) – used with red light for non-hyperkeratotic actinic keratoses (AK), Bowen's disease (BD), superficial BCC, and nodular BCC, though approvals vary by country.
- **5-ALA patch** – Alacare® (Spirig AG, Egerkingen, Switzerland) – approved for mild AK in a single session with red light, without lesion preparation.
- **BF-200 ALA** – Ameluz® (Biofrontera AG, Leverkusen, Germany) – licensed for use with red light in PDT.

Another formulation, **Levulan** (5-ALA; DUSA Pharmaceuticals, Wilmington, MA, USA), is approved in North America and some other regions for AK, using a blue light activation protocol. Many early studies used non-standardized, hospital-prepared formulations of aminolaevulinic acid (ALA), which limits direct comparison between study outcomes.

Photosensitizers

Topical photodynamic therapy, first described over two decades ago, involves the application of a photosensitizer prodrug—typically aminolaevulinic acid (ALA) or its methyl ester MAL. These compounds are metabolized via the heme biosynthesis pathway into protoporphyrin IX (PpIX), which, when activated by light of an appropriate wavelength, generates reactive oxygen species—particularly singlet oxygen—leading to apoptosis and necrosis of targeted cells.

Selective accumulation in lesional tissue is attributed to altered permeability of the stratum corneum and differences in porphyrin metabolism in diseased cells. Photosensitizer uptake may also occur in activated lymphocytes within the lesion.

ALA is hydrophilic, while MAL is more lipophilic, potentially allowing deeper penetration into skin lesions. However, comparative studies in AK and nodular BCC have not consistently shown superior efficacy of one agent over the other.

Recent data show that **BF-200 ALA**, a nanoemulsion formulation designed to enhance stability and skin penetration, achieved higher complete clearance rates than MAL in the treatment of thin to moderate facial and scalp AK: **78% vs. 64%**, respectively.

A self-adhesive, skin-colored 5-ALA patch applied directly to AK lesions—without crust removal—demonstrated superior efficacy compared to cryotherapy in clearing mild to moderate AK.

Techniques such as iontophoresis or chemical enhancers to improve photosensitizer penetration may increase PDT efficacy but remain experimental and are not yet part of routine clinical practice.